SG No. 10 C_{2h}^{1} $P2/m$ (b-axis setting) [monoclinic]

* plus set: $+[0, 0, 0]$

* Wyckoff site: `1a`, site symmetry: `2/m`

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, 0, 0]$	`[1,-2,-3,4]`

Table 2: Wyckoff bond: `1b@1a`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[0, 0, 0]$	`[1,2,-3,-4]`

Table 3: Wyckoff bond: `2c@1a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-3]`
2	$[-X, Y, -Z]$	$[0, 0, 0]$	`[2,-4]`

* Wyckoff site: `1b`, site symmetry: `2/m`

Table 4: Wyckoff bond: `1a@1b`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-2,-3,4]`

Table 5: Wyckoff bond: `1b@1b`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,-3,-4]`

Table 6: Wyckoff bond: `2c@1b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-4]`

* Wyckoff site: `1c`, site symmetry: `2/m`

Table 7: Wyckoff bond: `1a@1c`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-2,-3,4]`

Table 8: Wyckoff bond: `1b@1c`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,-3,-4]`

Table 9: Wyckoff bond: `2c@1c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[2,-4]`

* Wyckoff site: `1d`, site symmetry: `2/m`

Table 10: Wyckoff bond: `1a@1d`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-2,-3,4]`

Table 11: Wyckoff bond: `1b@1d`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,-3,-4]`

Table 12: Wyckoff bond: `2c@1d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2,-4]`

* Wyckoff site: `1e`, site symmetry: `2/m`

Table 13: Wyckoff bond: `1a@1e`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-2,-3,4]`

Table 14: Wyckoff bond: `1b@1e`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,-3,-4]`

Table 15: Wyckoff bond: `2c@1e`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-4]`

* Wyckoff site: `1f`, site symmetry: `2/m`

Table 16: Wyckoff bond: `1a@1f`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-3,4]`

Table 17: Wyckoff bond: `1b@1f`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,-3,-4]`

Table 18: Wyckoff bond: `2c@1f`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-4]`

* Wyckoff site: `1g`, site symmetry: `2/m`

Table 19: Wyckoff bond: `1a@1g`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-2,-3,4]`

Table 20: Wyckoff bond: `1b@1g`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,2,-3,-4]`

Table 21: Wyckoff bond: `2c@1g`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,-4]`

* Wyckoff site: `1h`, site symmetry: `2/m`

Table 22: Wyckoff bond: `1a@1h`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-3,4]`

Table 23: Wyckoff bond: `1b@1h`, bond symmetry: `2/m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,-3,-4]`

Table 24: Wyckoff bond: `2c@1h`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-3]`
2	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-4]`

* Wyckoff site: `2i`, site symmetry: `2`

Table 25: Wyckoff bond: `2a@2i`, bond symmetry: `2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, y, 0]$	`[1,-2]`
2	$[-X, 0, -Z]$	$[0, -y, 0]$	`[3,-4]`

Table 26: Wyckoff bond: `2b@2i`, bond symmetry: `2`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[0, y, 0]$	`[1,2]`
2	$[0, -Y, 0]$	$[0, -y, 0]$	`[3,4]`

Table 27: Wyckoff bond: `4c@2i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, 0]$	`[1]`
2	$[-X, Y, -Z]$	$[0, y, 0]$	`[2]`
3	$[-X, -Y, -Z]$	$[0, -y, 0]$	`[3]`
4	$[X, -Y, Z]$	$[0, -y, 0]$	`[4]`

* Wyckoff site: `2j`, site symmetry: `2`

Table 28: Wyckoff bond: `2a@2j`, bond symmetry: `2`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{2}, y, 0\right]$	`[1,-2]`

2	$[-X,\, 0,\, -Z]$	$\left[\frac{1}{2},\, -y,\, 0\right]$	`[3,-4]`

Table 29: Wyckoff bond: `2b@2j`, bond symmetry: `2`

No.	vector	center	mapping
1	$[0,\, Y,\, 0]$	$\left[\frac{1}{2},\, y,\, 0\right]$	`[1,2]`
2	$[0,\, -Y,\, 0]$	$\left[\frac{1}{2},\, -y,\, 0\right]$	`[3,4]`

Table 30: Wyckoff bond: `4c@2j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[\frac{1}{2},\, y,\, 0\right]$	`[1]`
2	$[-X,\, Y,\, -Z]$	$\left[\frac{1}{2},\, y,\, 0\right]$	`[2]`
3	$[-X,\, -Y,\, -Z]$	$\left[\frac{1}{2},\, -y,\, 0\right]$	`[3]`
4	$[X,\, -Y,\, Z]$	$\left[\frac{1}{2},\, -y,\, 0\right]$	`[4]`

* Wyckoff site: `2k`, site symmetry: `2`

Table 31: Wyckoff bond: `2a@2k`, bond symmetry: `2`

No.	vector	center	mapping
1	$[X,\, 0,\, Z]$	$\left[0,\, y,\, \frac{1}{2}\right]$	`[1,-2]`
2	$[-X,\, 0,\, -Z]$	$\left[0,\, -y,\, \frac{1}{2}\right]$	`[3,-4]`

Table 32: Wyckoff bond: `2b@2k`, bond symmetry: `2`

No.	vector	center	mapping
1	$[0,\, Y,\, 0]$	$\left[0,\, y,\, \frac{1}{2}\right]$	`[1,2]`
2	$[0,\, -Y,\, 0]$	$\left[0,\, -y,\, \frac{1}{2}\right]$	`[3,4]`

Table 33: Wyckoff bond: `4c@2k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[0,\, y,\, \frac{1}{2}\right]$	`[1]`
2	$[-X,\, Y,\, -Z]$	$\left[0,\, y,\, \frac{1}{2}\right]$	`[2]`
3	$[-X,\, -Y,\, -Z]$	$\left[0,\, -y,\, \frac{1}{2}\right]$	`[3]`
4	$[X,\, -Y,\, Z]$	$\left[0,\, -y,\, \frac{1}{2}\right]$	`[4]`

* Wyckoff site: `2l`, site symmetry: `2`

Table 34: Wyckoff bond: `2a@2l`, bond symmetry: `2`

No.	vector	center	mapping
1	$[X,\, 0,\, Z]$	$\left[\frac{1}{2},\, y,\, \frac{1}{2}\right]$	`[1,-2]`

continued ...

Table 34

No.	vector	center	mapping
2	$[-X, 0, -Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[3,-4]`

Table 35: Wyckoff bond: `2b@2l`, bond symmetry: `2`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[1,2]`
2	$[0, -Y, 0]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[3,4]`

Table 36: Wyckoff bond: `4c@2l`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[1]`
2	$[-X, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[2]`
3	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[4]`

* Wyckoff site: `2m`, site symmetry: `m`

Table 37: Wyckoff bond: `2a@2m`, bond symmetry: `m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[x, 0, z]$	`[1,4]`
2	$[-X, 0, -Z]$	$[-x, 0, -z]$	`[2,3]`

Table 38: Wyckoff bond: `2b@2m`, bond symmetry: `m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$[x, 0, z]$	`[1,-4]`
2	$[0, Y, 0]$	$[-x, 0, -z]$	`[2,-3]`

Table 39: Wyckoff bond: `4c@2m`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, z]$	`[1]`
2	$[-X, Y, -Z]$	$[-x, 0, -z]$	`[2]`
3	$[-X, -Y, -Z]$	$[-x, 0, -z]$	`[3]`
4	$[X, -Y, Z]$	$[x, 0, z]$	`[4]`

* Wyckoff site: `2n`, site symmetry: `m`

Table 40: Wyckoff bond: `2a@2n`, bond symmetry: `m`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[x, \frac{1}{2}, z\right]$	`[1,4]`
2	$[-X, 0, -Z]$	$\left[-x, \frac{1}{2}, -z\right]$	`[2,3]`

Table 41: Wyckoff bond: `2b@2n`, bond symmetry: `m`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[x, \frac{1}{2}, z\right]$	`[1,-4]`
2	$[0, Y, 0]$	$\left[-x, \frac{1}{2}, -z\right]$	`[2,-3]`

Table 42: Wyckoff bond: `4c@2n`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{2}, z\right]$	`[1]`
2	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, -z\right]$	`[2]`
3	$[-X, -Y, -Z]$	$\left[-x, \frac{1}{2}, -z\right]$	`[3]`
4	$[X, -Y, Z]$	$\left[x, \frac{1}{2}, z\right]$	`[4]`

* Wyckoff site: `4o`, site symmetry: `1`

Table 43: Wyckoff bond: `4a@4o`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, Y, -Z]$	$[-x, y, -z]$	`[2]`
3	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[3]`
4	$[X, -Y, Z]$	$[x, -y, z]$	`[4]`